

Arquitectura objetivo

Estructura de carpetas propuesta

AllRight/

```
|— .github/
|   └─ workflows/
|       └─ ci.yml                # Lint, test, build, docker
|       └─ cd.yml                # Deploy agnóstico (compose / registry)
|
|— apps/
|   └─ api/                      # NestJS bootstrap (capa Presentation)
|       └─ src/
|           └─ main.ts
|           └─ app.module.ts
|           └─ controllers/      # HTTP adapters
|           └─ guards/
|           └─ filters/
|           └─ interceptors/
|           └─ metrics/          # prom-client registry + /metrics
|               └─ Dockerfile
|               └─ nest-cli.json
|               └─ tsconfig.json
|
|— packages/
|   └─ domain/                  # Capa Domain (sin dependencias externas)
|       └─ src/
|           └─ entities/        # User, AuditLog, GeofenceZone, LocationProof
|           └─ value-objects/   # Coordinates, ProofHash, Email
```

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| | | └─ ports/                # Interfaces (repositorios, ZKP, cache)
| | | | └─ user.repository.port.ts
| | | | └─ audit.repository.port.ts
| | | | └─ location-proof.port.ts
| | | | └─ cache.port.ts
| | | └─ services/            # Lógica de dominio pura
| | | | └─ geofence.domain-service.ts
| | | | └─ location-privacy.domain-service.ts
| | └─ package.json
| |
| └─ application/            # Capa Application (use cases)
| | └─ src/
| | | └─ use-cases/
| | | | └─ auth/
| | | | | └─ login.use-case.ts
| | | | | └─ register-user.use-case.ts
| | | | └─ location/
| | | | | └─ update-location.use-case.ts
| | | | | └─ verify-location-proof.use-case.ts
| | | | | └─ generate-location-proof.use-case.ts
| | | | └─ audit/
| | | | | └─ log-action.use-case.ts
| | | └─ dto/
| | | └─ tokens.ts           # DI tokens para ports
| | └─ package.json
| |
| └─ infrastructure/        # Capa Infrastructure (adapters)
| | └─ src/

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|   |   └─ persistence/
|   |   |   └─ typeorm/          # Entities ORM + mappers
|   |   |   └─ repositories/     # Implementan ports del domain
|   |   └─ cache/
|   |   |   └─ redis/            # Sessions, rate-limit, GEO, concurrent users
|   |   └─ zkp/
|   |   |   └─ circuits/         # circom (.circom)
|   |   |   └─ wasm/             # Compilados (gitignored, CI build)
|   |   |   └─ snarkjs.adapter.ts
|   |   └─ observability/
|   |   |   └─ prometheus.module.ts
|   |   |   └─ metrics.service.ts
|   |   └─ config/
|   |       └─ env.validation.ts
|   └─ package.json
|
└─ web/                                # Frontend (evolución de /public)
    └─ public/
    └─ src/
|
└─ e2e/                                # Playwright (recomendado)
    └─ playwright.config.ts
    └─ fixtures/
    └─ tests/
        └─ auth.mobile.spec.ts
        └─ location.mobile.spec.ts
|
└─ infra/

```

```
|   └─ docker/
|   |   └─ docker-compose.yml          # Local dev
|   |   └─ docker-compose.prod.yml     # Producción
|   |   └─ .env.example
|   └─ prometheus/
|       └─ prometheus.yml
|   └─ grafana/
|       └─ provisioning/
|           └─ datasources/prometheus.yml
|           └─ dashboards/dashboard.yml
|       └─ dashboards/
|           └─ allright-overview.json
|
└─ package.json                        # Monorepo root (npm workspaces / pnpm)
└─ turbo.json                          # Optional: build pipeline
└─ README.md
```